

Semester Project

Traffic Flow Dynamics: Multivariable Modeling and Analysis

Objective

Multivariable calculus plays a vital role in analyzing and predicting complex systems such as traffic behavior in urban environments. In this project, you are given a dataset containing **30 days of traffic data** (or you can generate one from using AI) from major cities. The dataset includes:

- Time (t)
- Vehicle density (d) (vehicles/km)
- Average speed (v) (km/h)
- Traffic flow rate (F) (vehicles/hour)

You are required to model a function:

$$F = f(t, d, v)$$

You may use computational tools such as Python libraries including Scikit-learn to obtain the best-fit model. In real-world scenarios, traffic flow depends on many factors, but for this project, these three variables are sufficient. You should also understand how modeling tools work and how they approximate real-world behavior.

1. Predict Maximum and Minimum Traffic Flow

Use your modeled function to determine the combinations of time, density, and speed that result in:

- Maximum traffic flow (peak efficiency)**
- Minimum traffic flow (congestion or breakdown)**

Explain which variable (time, density, or speed) has the strongest influence on traffic flow.

2. Compute the gradient vector $\nabla T(t, h, a)$ at a given point (Programming Optional)

Compute the gradient vector $\nabla F(t, d, v)$, interpret the results and explain:

- Which variable is influencing the traffic flow the most?
- In which direction (combination of time, density, and speed) does traffic flow increase the fastest?

3. Steepest Descent Interpretation (Using Programming)

Find the negative of the gradient vector and

- In which direction does traffic flow decrease the fastest?
- What does this represent in real-world terms (e.g., congestion buildup)?

4. Directional Derivative Interpretation

- Choose a custom direction (e.g., increasing density while decreasing speed)

Compute:

$$D_{\mathbf{u}}F = \nabla F \cdot \mathbf{u}$$

Explain:

- How quickly traffic flow changes in that direction
- What does this mean in terms of real traffic conditions.

5. Compute Triple Integral Over a Region (Using Programming)

Choose a region in space, for example: $t \in [6,10]$, $d \in [20,80]$, $v \in [30,80]$ and compute the triple integral of $F(t, d, v)$ over this region.

Interpret: This represents the **total accumulated traffic flow** over a time period and conditions. Explain its significance (e.g., total vehicles passing through a system).

6. Identify the Hottest/Coldest City

Since the dataset includes multiple cities:

- Apply your model separately for each city
- Determine:
 - The city with the **highest average traffic flow (most efficient)**
 - The city with the **lowest traffic flow (most congested)**.

Report Requirements

Students are required to submit a complete report of the project prepared in MS Word in their own words. A report is written in third person, i.e., I, We, Us, are not used for writing the report. The headings of the report should be the same as the heading and subheadings of the tasks given above. The report should also include an introduction section describing the task, objectives, and methodology. Include a flowchart in this section to explain the methodology. Do not just paste the results or screenshots, also meaningfully analyze the results. The report should also include a conclusions section and a contributions section. Provide any references used in the report at the end. Attach complete codes as annexure.

Each page should be numbered. The report should be written in the Times New Roman typeface only. The size of the font should be 12. The size of first and second level of headings should be 14 bold, and 12 bold, respectively. The alignment of the report should be justified, while pictures and tables should be center aligned with relevant captions. The option to align the text left, right, center, and justify can be found under paragraph options on *Home* tab. Line and paragraph spacing should be set as 1.5. Optimally utilize the available space on each page, do not leave blank space on a page unnecessarily.

Guidelines

This project is an open-ended problem designed to demonstrate the application of Multivariable Calculus in real life in general and Computer Science in particular. The open-ended nature of the problem means that this problem can be solved in more than one way using various techniques and methodologies. You are free to adopt any technique and solution methodology to solve this problem. If you have any confusion, you can discuss your query via email. Project guidelines are summarized below:

- This is a group project and carries 70 marks.
- A group can have a maximum of 3 students. One of the aims of this project is to enable students to work effectively in a team. **Therefore, this project cannot be done individually.** The project can be done in pairs.
- Plagiarized work (from internet, AI, or fellow students) will result in zero marks. Similarity index of less than 18% and AI content less than 15% is allowed.
- The deadline for complete project submission on [google classroom](#) ([one MS Word file and one pdf of the same Word file including all the codes and by-hand solutions](#)) is **Thursday 07 May 2026 before 11:59pm**. Do not submit your project in a .zip or .rar format. You can submit additional files; however, the single PDF and MS Word file must also include all the information such as codes and figures.
- Name of your project report file must be as per following format: **ID1_ID2_ID3_MT1008_Project_Section.**
- Do not submit your project via email, it will not be considered.
- Late submissions will not be considered.